Bio-IFGT: Biological Morphogenesis as Information-Flow Attractor Dynamics

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Abstract

This paper presents a minimal common framework for describing biological systems in terms of information density I, information flow $\mathbf{J}$, and an induced potential Φ. These are source-local formal or model quantities with proposed biological readings, not operationally identified observables. In this framework, DNA is represented abstractly through a parameter set θ that constrains system dynamics, rather than as an encoded final form; no concrete genotype-to-parameter map is supplied.

On this basis, development and morphology are given a proposed attractor-based interpretation, and life is described through a candidate sustained non-equilibrium region. The existence, stability, and operational biological identification of such regions remain separate questions. The probability of the emergence of life is correspondingly reformulated as a region-entry target rather than a single-point target; no numerical probability is calculated here.

Intelligence is treated only as a forward conceptual structure: a proposed second attractor supported by the life-attractor description. The paper does not introduce a new biological mechanism. It provides a common structural description using a minimal set of dynamical quantities; the present simulation is a qualitative model-local witness and is not tested here against an external biological dataset.

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1 Introduction

Many biological phenomena are observed as development, morphogenesis, differentiation, and evolution. These phenomena are often described using language such as design, program, or genetic instruction. Such language can be useful at an intuitive level, but its relation to the underlying physical and chemical processes is not always explicit.

DNA sequences constrain molecular production and reaction pathways. They do not directly specify a completed structure, purpose, or intention. The fact that cells with the same DNA can differentiate into distinct states illustrates this point. DNA is therefore represented here abstractly through a parameter set θ that constrains the dynamics of the system; a concrete genotype-to-parameter map and parameter provenance are not supplied.

This paper describes biological systems as processes of state change and sustained structure. We use three source-local formal quantities: information density I, information flow $\mathbf{J}$, and induced potential Φ. They provide a common model description of development and morphogenesis, while their operational identification with biological observables remains separate.

The central feature of the framework is that it does not require design or purpose as explanatory primitives. Form and function are described as possible outcomes of state change under an attractor-based interpretation; convergence and stability in a concrete biological system are not established by the generic relations alone.

The aim is to describe existing phenomena in a consistent form. We combine a minimal set of variables, a continuity equation, and an attractor-based interpretation of stability.

Section 2 gives the intuitive picture. Section 3 defines the formal mapping. Section 4 introduces the core equations. Sections 5 and following apply the framework to biological phenomena, simulation, and scope.

This paper builds on the Information-Field-Geometry-Theory (IFGT), which formulates the motion of systems toward closure and the persistence of quasi-closure structures. In IFGT, the degree of closure C characterizes how far a system has progressed toward a stable configuration. In biological systems, the field I should not be identified directly with the IFGT quantity $I = f(1 - C)$. It is a source-local, coarse-grained constraint-density field used to describe local structural constraint, differentiation, and retained bias within a particular observational window; no operational biological measurement map is supplied here. The relation is therefore a proposed correspondence between descriptive layers, not an identity, derivation, or typed translation between variables. The sustained non-equilibrium structure of living systems may likewise be interpreted, in IFGT terms, through quasi-closure. This proposed structural reading does not establish a biological mechanism or complete any upstream VED/IFGT bridge.

2 Intuitive Picture

This section gives the informal picture behind the formalism. The aim is not to define the framework rigorously, but to indicate what is being described.

2.1 State and Change

A biological system is not treated as a fixed object. It is treated as a process in which states continue to change. Molecules are produced, degraded, transported, and coupled through reactions. Cellular states change over time.

The important point is that change does not stop, while the system can still preserve an overall organization. In the intuitive terminology used here, biological stability is read not as static

immobility but as sustained organization under continuing update; no maintenance mechanism or stability result is established by this statement.

2.2 Bias and Flow

State change is not uniform. Some regions produce particular molecules more strongly. Other regions suppress them. These asymmetries give direction to state change. We call this directionality a flow.

The flow arises from production, diffusion, transport, and interaction. It is not a separate substance. It is the net direction and rate of state update.

2.3 Stable Regions

When flows exist, not all states behave in the same way. Some states are maintained more easily, while others decay or are displaced. As a result, regions can appear to which the system repeatedly returns.

Such regions are stable in the dynamical sense. They can absorb nearby deviations and preserve a recognizable structure.

This is a conditional intuitive description: whether a concrete region exists and has the stated return property must be established independently.

2.4 Appearance of Form

Development and morphogenesis are interpreted through candidate persistent configurations under continued state change. A form is not assumed in advance. This is an intuitive attractor-based reading, not an established convergence or biological-stability result.

Cellular distributions and organ-level patterns may be described through such arrangements at the interpretive level.

2.5 Without Design Assumptions

This view does not require purpose, design, or instruction as primitive concepts. It requires only state change, asymmetry, flow, and stable configurations.

2.6 Summary

The intuitive picture is that biological systems can be understood through state change, flow, and stable structure. The formal sections describe this picture using information density I, information flow $\mathbf{J}$, and induced potential Φ. Here "stable" and "return" are intuitive, conditional terms. They do not establish an invariant basin, asymptotic convergence, biological robustness, or regeneration.

3 Formal Mapping

This section maps the intuitive picture onto formal quantities. The goal is to make the correspondence between intuition and equations explicit, not to derive a complete model for any particular organism.

Within this continuum representation, the field expressions below declare states and domains; the potential and flow formulas are constitutive or effective relations; and the internal-time formula is an accumulation definition relative to t. These roles do not make the same quantities independently integrated variables of the finite toy code.

3.1 State Description

A biological system is represented as a spatially and temporally varying state. For position $\mathbf{r} \in \mathbb{R}^3$ and time $t \in \mathbb{R}$, we consider coarse-grained fields rather than all microscopic molecular details.

The notation in this section is source-local to the Bio-IFGT continuum description. In particular, I and $\mathbf{J}$ are not identified by their glyphs with the general IFGT field, the discrete VED current J_{ij}, or the finite toy-model arrays. The toy simulation does not directly integrate independent $\mathbf{J}$ or Φ arrays.

3.2 Information Density

The bias or constraint present in a local state is represented by the information density

$$I(\mathbf{r}, t). \tag{1}$$

Here, information does not mean semantic content. Within this representation it denotes the degree of local structure, constraint, differentiation, or interdependence. This is a source-local model definition and proposed biological descriptor, not an operationally calibrated biological observable.

3.3 Information Flow

The direction and rate of state update are represented by the information flow

$$\mathbf{J}(\mathbf{r}, t). \tag{2}$$

The flow is used as a proposed aggregate description of contributions from reactions, diffusion, transport, and interactions. It describes how the coarse-grained state is updated; no event-to-current aggregation or measured-flux map is supplied here.

3.4 Induced Potential

When information density is spatially biased, state change becomes asymmetric. This asymmetry is represented by an induced potential

$$\Phi = \Phi[I]. \tag{3}$$

This represents an effective potential induced by the distribution of I. It is not identified here with a measured biochemical potential or a demonstrated molecular restoring mechanism. The corresponding gradient-driven bias is written as

$$\mathbf{F} = -\nabla\Phi. \tag{4}$$

Here the bold vector $\mathbf{F}$ is distinct from the scalar response function F_k used for threshold conditions and the state-space evolution field $F(x;\theta)$.

3.5 Relation to Flow

The information flow can be written as the sum of a diffusive component and a component induced by the potential:

$$\mathbf{J} = -D\nabla I + \mu\mathbf{F}, \tag{5}$$

where D is a diffusion coefficient and μ is a response coefficient.

3.6 Treatment of Time

The equations are written using an external time parameter t. This is the time with respect to which state change is recorded. We may also introduce an internal time τ that measures the progress of state update:

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \rho(I, \mathbf{J}), \tag{6}$$

where ρ is a state-dependent progression rate. This Bio-local progression functional is not the VED node-density quantity or an upstream continuum density. In this paper, equations of motion are written in t, while τ is used interpretively. Neither variable is identified with the finite simulation loop step or its `DT`; no map to VED or IFGT time variables is supplied here.

3.7 Summary

The mapping is

$$\text{bias or constraint} \mapsto I, \tag{7}$$

$$\text{state update} \mapsto \mathbf{J}, \tag{8}$$

$$\text{asymmetry in update} \mapsto \Phi. \tag{9}$$

With these quantities, the intuitive picture becomes a continuous-field description. The mapping is formal within the stated representation; operational biological identification and cross-scale translation remain separate open tasks.

4 Core Equations

This section introduces the minimal equation system used throughout the paper. The equations are not tied to a specific biological system. They provide a general form for describing state change, flow, and candidate persistent structure.

Formally, I and $\mathbf{J}$ are field declarations in the stated continuum domain, Equation (12) is a balance relation, and Equations (13)–(15) are constitutive or effective relations whose use is conditional on the chosen $\Phi[I]$ and differentiable spatial representation. This system is not a line-by-line finite update algorithm for the toy simulation.

4.1 Basic Variables

The basic fields are the information density

$$I(\mathbf{r}, t) \tag{10}$$

and the information flow

$$\mathbf{J}(\mathbf{r}, t). \tag{11}$$

The field I represents local density of structure or constraint within the model. The field $\mathbf{J}$ represents the direction and rate of model-state change. Neither field is, by this definition alone, an operationally measured biological observable or flux.

4.2 Continuity Equation

Information density obeys the continuity equation

$$\frac{\partial I}{\partial t} + \nabla \cdot \mathbf{J} = \sigma, \tag{12}$$

where $\sigma(\mathbf{r}, t)$ is a local generation rate. It is the Bio-IFGT source slot for this balance equation, not an identity with VED or IFGT uses of σ, nor with the toy morphogen source σ_M. A gene-related reading is a source-local interpretation of this slot; no independently calibrated gene-production map is supplied. This equation records local production, outflow, and conservation-like transport of the coarse-grained quantity I.

4.3 Structure of the Flow

The flow is written as

$$\mathbf{J} = -D\nabla I + \mu \mathbf{F}, \tag{13}$$

where D is a diffusion coefficient, μ is a response coefficient, and $\mathbf{F}$ is an induced direction field.

4.4 Potential

The direction field is determined by the induced potential:

$$\mathbf{F} = -\nabla \Phi. \tag{14}$$

The potential is a functional of information density:

$$\Phi = \Phi[I]. \tag{15}$$

The specific form of Φ is system-dependent and left unspecified. Thus, biases in information density induce a gradient-driven bias in state change. The sign convention is fixed only after $\Phi[I]$ is specified. For example, choosing $\Phi \sim -I$ makes the induced component $-\nabla\Phi$ point toward increasing constraint density. Other proposed biological applications may require the opposite convention or additional response terms; the present relation is not itself an operational observable map.

4.5 Dynamical Loop

Equations (12)–(15) form a loop: the distribution of I induces Φ, the potential shapes $\mathbf{J}$, and the flow redistributes I. In general this loop does not necessarily terminate in a static equilibrium. It can produce non-equilibrium structures that are sustained under continued update.

4.6 Two-Layer Description of Time

The equations use the external time parameter t. An internal time τ can be introduced as

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \rho(I, \mathbf{J}), \tag{16}$$

where $\rho(I, \mathbf{J})$ is a state-dependent progression rate. The variable t is used for equations of motion. The variable τ is used to interpret the internal progress of the system.

4.7 Stable Structures

Under the terminology used here, a region of state space is called stable if states within or near it tend to remain there or return after a stated class of perturbations. Such a region may be interpreted as an attractor. This is a conditional definition: the equations alone do not establish existence, an invariant basin, asymptotic convergence, or general bidirectional return for a concrete system.

4.8 Summary

The minimal equation system is

$$\frac{\partial I}{\partial t} + \nabla \cdot \mathbf{J} = \sigma, \tag{17}$$

$$\mathbf{J} = -D\nabla I + \mu \mathbf{F}, \tag{18}$$

$$\mathbf{F} = -\nabla \Phi, \tag{19}$$

$$\Phi = \Phi[I]. \tag{20}$$

This system provides a common form in which state change, flow formation, and candidate persistent structures can be described. It does not by itself prove their stability.

5 Gene × Information Flow

This section describes DNA and gene-related processes in terms of information density and information flow. The aim is to describe the emergence of development, form, and cell fate as sequences of state change, rather than as the execution of semantic instructions.

5.1 Role of DNA

A DNA sequence s is written as

$$s \in \{A, T, G, C\}^N. \tag{21}$$

The sequence has no intrinsic meaning or purpose in the present formalism. It constrains molecular possibilities, reaction propensities, and threshold conditions. Here N denotes sequence length; it is not the neural-commitment field N of the later toy simulation. Likewise the nucleotide letters are alphabet symbols, not the later state labels or toy fields that reuse a glyph.

Thus DNA is represented at this abstract model layer through a parameter set

$$\theta = \theta(s), \tag{22}$$

which constrains system dynamics. It is not treated as a direct specification of a completed form, and the displayed relation does not supply a concrete genotype-to-parameter implementation or parameter provenance.

5.2 Gene Events as State Changes

Transcription, translation, binding, and modification are represented as local operations that change information density:

$$I(\mathbf{r}, t) \mapsto I(\mathbf{r}, t) + \Delta I. \tag{23}$$

Although these events are discrete, their aggregate effect may be modeled through a continuous information flow $\mathbf{J}(\mathbf{r}, t)$. This is a proposed aggregation and process interpretation: no explicit event-to-current map is supplied. A gene network may accordingly be read as a candidate source of such flow, rather than being identified with $\mathbf{J}$.

5.3 Connection to the Continuity Equation

In the continuity equation

$$\frac{\partial I}{\partial t} + \nabla \cdot \mathbf{J} = \sigma, \tag{24}$$

the source term σ is the formal source slot and may be given a gene-related production interpretation. The flow $\mathbf{J}$ may similarly describe the aggregate effect of such events across space and time. Neither reading is an operational identity or a measured biological map in the present paper.

5.4 Bias and Induced Potential

When information density becomes biased, it induces a potential

$$\Phi = \Phi[I]. \tag{25}$$

State change is then biased according to

$$\mathbf{F} = -\nabla\Phi. \tag{26}$$

This is not a new fundamental force. Within a specified attractor-based model, it can represent an effective restoring bias toward a selected region. It is not identified with a measured biochemical potential or demonstrated molecular mechanism, and the existence of a stable basin or general return is not supplied by this formal relation alone.

5.5 If–Then Conditions as Thresholds

The conditional effects associated with DNA are implemented as threshold conditions in a continuous field. For a response function F_k, a reaction may switch when

$$F_k(I, \mathbf{J}, \mathbf{r}, t; \theta) > \theta_k. \tag{27}$$

Here θ_k is a parameter associated, at the abstract model layer, with the DNA-conditioned parameter set θ. The condition is not a symbolic rule in isolation. It is a threshold in the dynamics; a concrete sequence-to-threshold map is not implemented here. The response function F_k is not the vector bias $\mathbf{F}$ and is not the state-space evolution field $F(x; \theta)$.

5.6 Interpretation of Form

The resulting picture is as follows. DNA supplies parameters. Information density and information flow are formed. Biases in I induce Φ. Flow is shaped by the induced potential. Thresholds branch the state. Resulting configurations may then be read through an attractor-based interpretation. Their identification with observed cell fate or morphology is a separate biological correspondence, not a consequence established by this sequence alone.

5.7 Proposed Cross-Scale Reuse

The same descriptive vocabulary may be reused, by proposal, at molecular, cellular, and tissue scales. The level of coarse-graining and the source-local objects change. No typed scale map, coarse-graining operator, preserved invariant, or theorem of role identity is supplied here; the cross-scale relation is a formal analogy rather than an established identity or universality result.

5.8 Summary

DNA is treated as a condition-setting parameter set. The framework uses information density and information flow to describe candidate persistent configurations; a biological structure reading is interpretive. This provides the basis for the treatment of development and morphogenesis in the next section.

5.9 Origin of the Parameter Set

At this descriptive layer, the parameter set θ need not be treated as a conceptual primitive. In IFGT terms, it may be read, by proposed analogy, as a quasi-closure structure retained through prior dynamics of difference. DNA sequences may accordingly be interpreted as historically stabilized constraint patterns. This reading supplies neither parameter provenance nor a concrete genotype-to-θ mechanism for a given organism, and it is not a typed IFGT/Bio translation.

6 Development and Morphogenesis

This section applies the preceding framework to development and morphogenesis. Development is proposed to be read not as execution of a design, but through state changes that may approach persistent configurations. The generic relations do not by themselves establish biological convergence.

6.1 Initial State and Asymmetry

A fertilized egg is an initial state with broad potential in state space. At this stage, information density may be approximately uniform, and no macroscopic structure is yet fixed.

Small asymmetries nevertheless exist through initial conditions and environmental fluctuations. These asymmetries can seed later structure formation.

6.2 Morphogens as Information Density

During development, morphogens form spatial concentration profiles. In the present framework, such profiles may be represented through the source-local information-density field:

$$I(\mathbf{r}, t). \tag{28}$$

Within that model representation, local production creates gradients in I, and diffusion and degradation shape the resulting distribution. This is a proposed qualitative biological correspondence, not an identity or calibrated measurement map between a morphogen and I.

6.3 Dynamics

A standard coarse-grained form for this process is

$$\frac{\partial I}{\partial t} = D\nabla^2 I - \lambda I + \sigma(\mathbf{r}, t; \theta), \tag{29}$$

where D is a diffusion coefficient, λ is a decay rate, and σ is a local source constrained by the parameter set θ.

This equation is not special to this framework. It is a standard reaction–diffusion form used here as a component of the information-flow description. It is a continuous coarse-grained evolution relation, not an assertion that the later finite toy code is an exact PDE solver or an identical dynamical object.

6.4 Induced Potential and Gradient

The distribution of I induces a potential

$$\Phi = \Phi[I]. \tag{30}$$

State change is biased by

$$\mathbf{F} = -\nabla\Phi. \tag{31}$$

As a result, model states become biased toward regions or gradients associated with I. This effective relation gives directionality within the chosen model; it is not a demonstrated biochemical restoring mechanism.

6.5 Differentiation by Thresholds

Model-state labels can branch according to thresholds in I, providing a proposed cell-fate analogy. For example,

$$I(\mathbf{r}) < \theta_1 \quad \Rightarrow \quad \text{state A}, \tag{32}$$

$$\theta_1 \leq I(\mathbf{r}) < \theta_2 \quad \Rightarrow \quad \text{state B}, \tag{33}$$

$$I(\mathbf{r}) \geq \theta_2 \quad \Rightarrow \quad \text{state C}. \tag{34}$$

The variable I is continuous, while the model output labels are discrete. Biological differentiation may therefore be interpreted through a threshold response to continuous state change, but an observed-morphology map is not supplied. Mathematically, the displayed implications are threshold classification rules; their state labels and biological reading are a separate interpretive layer.

6.6 Morphology as Attractor

The combination of dynamics and thresholds can yield persistent model distributions and differentiation labels under specified conditions. Reading such a configuration as observed developmental morphology is a separate biological interpretation. Neither a stationary attractor nor empirical morphological convergence follows from the displayed relations alone.

6.7 Robustness

Development is often described as reproducible and robust. In an attractor-based interpretation, convergence of multiple nearby initial states to the same region would provide a candidate account of robustness. The present formal relations and finite toy model establish neither biological robustness nor arbitrary-perturbation return; they show only limited model-local persistence, relaxation, and recommitment over tested intervals.

6.8 Summary

Development and morphogenesis are described through small initial asymmetries, distributions of information density, gradient-driven bias, threshold branching, and a proposed attractor-based interpretation. The finite construction supports model-local spatial organization and qualitative analogy; biological mechanism and operational morphology correspondence remain separate. No design or purpose is required as an additional explanatory primitive.

7 Life as Attractor

This section introduces a proposed life-attractor terminology: life is considered not as a completed object, but through sustained motion in a bounded state-space region. This conceptual view places maintenance, development, differentiation, and regeneration in one dynamical vocabulary without establishing their biological mechanisms.

7.1 State-Space Description

A biological system is represented by a state vector

$$x(t) = (x_1(t), x_2(t), \ldots, x_n(t)), \tag{35}$$

where the components may include model representations of molecular concentrations, expression levels, reaction rates, and local environmental variables. Their operational measurement and calibration are not specified by this declaration.

The system evolves as

$$\frac{\mathrm{d}x}{\mathrm{d}t} = F(x;\theta), \tag{36}$$

where θ is the abstract parameter set associated with DNA-conditioned model terms such as reaction propensities and thresholds; a concrete genotype-to-parameter map is not supplied. Here x is a state vector rather than a spatial coordinate, and $F(x;\theta)$ is its evolution field; this F is distinct from both the continuum bias vector $\mathbf{F}$ and the threshold response F_k. The first display in this subsection is a state/domain declaration and the second is a generic evolution rule. Neither specifies the ordered finite updates of the developmental toy code.

7.2 Internal Time

In addition to external time t, we introduce an internal time τ that measures the progress of state update:

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \rho(x). \tag{37}$$

The function $\rho(x)$ is a state-dependent progression rate. It shares a progression role with the earlier $\rho(I, \mathbf{J})$, but no exact identity between the two functionals is specified; neither is an upstream density

variable. The variable τ represents accumulated state change or information update, and need not coincide with t. Equations of motion are written in t; τ is used to interpret their internal progression.

7.3 Definition of Life

Within this model terminology, life is represented not by a single state but by motion that continues while remaining within a bounded region of state space. Such motion is stipulated to be non-static, non-divergent, and constrained within a viable range, while τ continues to increase. An operational criterion that independently identifies biological life or the relevant system boundary is not supplied here.

7.4 Life as Attractor

Let Ω be the state space and let

$$\Omega_{\text{life}} \subset \Omega \tag{38}$$

be a region such that, once the system enters it, the dynamics tend to return to it after a stated class of moderate perturbations. We call Ω_{life} the life attractor within this conceptual model. This names a conditional region definition; it does not establish that such a region exists, is reachable or invariant in a concrete system, or is operationally identical to biological life. The finite developmental toy model is not demonstrated here to instantiate Ω_{life}.

The intended interpretation is that organization of this region would arise through interactions within the system and its environment rather than from a prescribed final endpoint. The state representation, variables, sources, parameters, boundaries, and viability conditions may nevertheless be supplied; endogenous construction of the region remains open.

7.5 Non-Equilibrium Sustained Structure

The proposed life attractor is not an equilibrium-state picture: matter may enter and leave, molecules may be replaced, and energy may be consumed while organization persists. This is a non-equilibrium maintenance interpretation. The present equations do not supply a dedicated resource budget or demonstrate a biological self-maintenance mechanism.

7.6 Relation to Development

The developmental process described in Section 6 may be interpreted as approach from an initial state toward a proposed life-attractor region. If multiple initial conditions returned to the same region under specified perturbations, that behavior could provide a model account of reproducibility and robustness; neither the general return property nor biological robustness is established here.

7.7 Differentiation and Sub-Attractors

Within the proposed life-attractor region, more specific conditionally stable regions may exist. These are termed sub-attractors. Cell differentiation may be interpreted as transition into such sub-attractors within Ω_{life}. This is a conceptual description of discreteness, plasticity, and partial reversibility, not an established biological identification or stability result.

7.8 Robustness and Regeneration

In an attractor-based interpretation, return after a specified perturbation would provide a candidate mechanism for robustness. The finite toy evidence distinguishes limited relaxation and monotonic recommitment from general bidirectional return. Regeneration remains a proposed reconstruction interpretation for larger disturbance; no dedicated reconstruction model, independent recovery criterion, or biological regeneration result is supplied here.

7.9 Summary

The proposed view treats life not merely as a collection of matter or a static structure, but as sustained motion in a bounded region of state space. Dynamics involving information density I, information flow $\mathbf{J}$, and induced potential Φ provide the descriptive language. Biological identification, stability, and maintenance remain distinct open requirements. The next section uses this view to revise the target of the emergence-probability question.

8 Probability Revision

This section revisits the formulation of the probability of the emergence of life. The aim is not to assign a numerical value, but to clarify what the probability is taken to measure.

8.1 Conventional Formulation

The emergence of life is often treated as the probability that a particular molecular structure appears by chance. In this view, many conditions must be satisfied simultaneously, and each may have low probability. The combined probability can therefore become very small.

This view is coherent if life is assumed to be a particular completed configuration. The present framework uses a different object of probability.

8.2 The Issue

Under the proposed terminology of Section 7, life is represented not by a fixed structure but by sustained motion in a bounded state-space region. This is a model target; its operational identification with biological life remains open.

The emergence of life is then interpreted as reaching this region and remaining within it, rather than hitting one specified structure.

8.3 Probability as Region Entry

For this conceptual reformulation, let the proposed life-attractor region be

$$\Omega_{\text{life}} \subset \Omega. \tag{39}$$

The probability question is then reformulated as the probability of reaching Ω_{life}. This is a region-entry target rather than an exact-point target; no sample space, measure, projection, or numerical probability is supplied.

8.4 Structure of Trials

In the schematic trial picture, molecular reactions and material cycles repeat as long as enabling conditions persist: non-persistent states disappear while persistent states remain. This motivates the region-entry reformulation; it is not a demonstrated biological maintenance mechanism or an independently measured selection process.

8.5 Dimensional Reduction

In a point-target view, many degrees of freedom must be specified simultaneously. The present framework proposes describing the relevant conditions through fewer effective variables that characterize entry into a candidate region. Without a projection and probability measure, this is a structural reformulation rather than an established quantitative dimensional reduction.

8.6 Interpretation

This is not a numerical reassessment of probability. It is a structural redefinition of the object being counted. Life is neither a miraculous hit nor an automatic necessity. It is a state that can appear when sustaining conditions exist.

In IFGT terms, this reformulation has a proposed structural interpretation: the Bio-IFGT life-region target may be compared with entry into and persistence within a quasi-closure regime. This is a correspondence of descriptive roles, not an identity, typed translation, or completed biological mechanism. The probability question may therefore be posed schematically as reaching and maintaining such a regime rather than assembling a fixed molecular configuration; no probability measure, empirical estimate, or IFGT/Bio bridge is supplied here.

8.7 Connection to Intelligence

This reformulation makes the next question more precise: after life is established, why do only some systems reach intelligence? Intelligence is not a simple continuation of life. It requires additional structural conditions. The next section treats intelligence as a second attractor.

9 Intelligence as a Second Attractor

This section presents a forward conceptual proposal in which intelligence is treated not as a direct extension of life, but as a second attractor supported by the proposed life-attractor description. This separates the candidate conditions for life from additional conditions considered for intelligence.

9.1 Position of Intelligence

Intelligence is often treated as an advanced function of living systems. Within the terminology proposed here, life is described through an attractor region, while many living systems persist without developing intelligence. Intelligence is therefore not modeled merely as a strengthening of that region.

9.2 Second Attractor

Intelligence appears when information is not only flowing, but also retained, referenced, and reused. Past states influence present states, and present states modify future trajectories. This creates recurrent loops of information flow.

Such loops are proposed to require conditions that differ from those considered for life alone. Intelligence is therefore interpreted as a candidate second attractor supported by, but not identical to, the life-attractor description.

9.3 Structural Independence

Formally, one may write

$$\Omega_{\text{intel}} \subset \Omega_{\text{life}}, \tag{40}$$

but the internal structure of Ω_{intel} is proposed to depend on additional constraints. Candidate conditions include persistent reference to state history, temporal accumulation of information, and sharing of information across multiple components. Their sufficiency and their biological or cognitive translation are not established in the present Bio model.

9.4 Probability Structure

Following Section 8, the emergence of intelligence is treated as the satisfaction of conditions distinct from the emergence of life. Its probability therefore has a conditional structure. A schematic factorization is

$$P_{\text{intel}} \approx P_{\text{life}}\, P_{\text{second|life}}, \tag{41}$$

where $P_{\text{second|life}}$ denotes the probability of reaching the second attractor given the life attractor. This expression is not a numerical estimate. It records the two-stage structure of the problem.

9.5 Non-Teleological Interpretation

Treating intelligence as a proposed second attractor does not give it a privileged status. An attractor is not a goal. Here it names a candidate structure that would persist when its conditions are satisfied. Intelligence is therefore neither inevitable nor a final form in this conceptual account.

9.6 Relation to Internal Time

In the notation of Section 7, life is a structure in which τ continues to progress. Intelligence is characterized by the reference and reuse of the history of τ. In this sense, intelligence is not merely flow. It is recursive information flow.

9.7 Summary

Intelligence is not treated as a simple extension of life. It is proposed as a recursive information-flow structure supported by the life-attractor description. This forward interpretation motivates additional conditions; it does not establish a stable second-attractor implementation or a cognitive translation.

9.8 Connection to Further Work

The next question is how this second attractor is implemented. Its realization may involve memory, temporal integration, language, and inter-individual sharing. These issues are beyond the minimal biological framework developed here.

10 Relation to Note

This section clarifies the proposed relation between this paper and the corresponding note-style exposition. The aim is not to summarize the theory, but to compare analogous roles in different descriptive layers.

10.1 Difference in Descriptive Layer

This paper is written as an L3 formal layer. The note essay is written as an L1 intuitive layer. They are proposed to concern a related target, but their purposes differ. The paper fixes definitions, equations, and relations. The note essay supports intuitive understanding through accessible language and metaphor. The exact L1 source/version locator is not fixed here.

The relevant relation is therefore a proposed structural correspondence, not identity of objects, dynamics, or wording and not an exact translation.

10.2 Correspondences

The concepts used in this paper correspond to intuitive expressions in the note essay:

$$I \leftrightarrow \text{local concentration, bias, or structured region,} \tag{42}$$

$$\mathbf{J} \leftrightarrow \text{flow, repetition, or ease of passage,} \tag{43}$$

$$\Phi \leftrightarrow \text{return tendency or stable place,} \tag{44}$$

$$\text{attractor} \leftrightarrow \text{returning to the same region or preserving form.} \tag{45}$$

These are not direct translations. They are candidate analogies between differently typed descriptions; no exact mapping or empirical identity of phenomena is established.

10.3 Development and Form

The treatment of development and morphogenesis in Section 6 corresponds to the note-level idea that repeated changes form flows and that flows stabilize into form. Both descriptions avoid taking design or purpose as primitives.

10.4 Definition of Life

Section 7 proposes a life-attractor terminology based on sustained motion in a bounded region. In the note essay, this appears as the idea that life is not an object but a state. The attractor reading is conditional; it does not supply a stability result or an operational biological life criterion.

10.5 Probability Revision

The probability revision in Section 8 corresponds to the note-level statement that life is not best understood as a miracle-like exact hit. The shared structure is the shift from a point target to a region, and from exact matching to arrival and persistence.

10.6 Connection to Intelligence

The proposed treatment of intelligence as a second attractor in Section 9 corresponds to a note-level description of a flow that appears to think. Neither source supplies a stable second-attractor implementation; the conceptual distinction between life and intelligence is retained.

10.7 Relation to Appendices

Intermediate appendices can bridge the note essay and this paper. They may introduce state, flow, and stability with minimal mathematics. Future versions can be updated to reflect the structure fixed here.

10.8 Summary

This paper and the note essay use different descriptive forms. The paper fixes the Bio-IFGT formal structure, while the note essay supplies an intuitive presentation. Their proposed correspondence can improve coherence across layers, but it is not an identity or a typed translation and remains dependent on fixing the exact L1 source/version.

11 Minimal Developmental Simulation (Bio-IFGT Toy Model)

11.1 Purpose

This section introduces a minimal Bio-IFGT toy model. The purpose is to illustrate how a small set of DNA-like parameters, including sources, diffusion, decay, thresholds, and response rates, can generate reproducible finite-horizon model patterns without encoding a completed output array. The prescribed source and positional window do, however, supply coarse spatial information.

11.2 Model Setup

We consider a two-dimensional domain representing an embryo-like field. The system evolves over discrete time steps and consists of four primary fields:

- morphogen-like model field $M(x, y, t)$,
- model information density $I(x, y, t)$,
- neural-like commitment $N(x, y, t)$,
- eye-like commitment $E(x, y, t)$.

An auxiliary boundary field is also introduced:

- gradient-sensitive boundary-like accumulation $B(x, y, t)$.

These symbols are source-local toy fields and become finite arrays in the executable implementation. The toy I is a finite proxy for the Bio-IFGT field, not a carrier identity with it, and the code does not directly integrate independent $\mathbf{J}$ or Φ arrays. The discrete loop step and its `DT` are runtime quantities, not the Bio internal variable τ or a biological clock.

11.3 Dynamics

Morphogen field. The morphogen field evolves by diffusion, decay, and localized production:

$$\frac{\partial M}{\partial t} = D_M \nabla^2 M - \lambda_M M + \sigma_M(x, y). \tag{46}$$

Information density. The information density accumulates from the morphogen field while also diffusing and decaying:

$$\frac{\partial I}{\partial t} = D_I \nabla^2 I - \lambda_I I + \alpha M. \tag{47}$$

Neural commitment. Neural commitment is updated by thresholding the information density:

$$N_{t+1} = N_t + \eta_N S(I - \theta_N)(1 - N_t). \tag{48}$$

Boundary field. The boundary field is derived from the gradient structure of information density:

$$G = |\nabla I|. \tag{49}$$

This displayed G is the paper's raw gradient magnitude. The current code normalizes that magnitude globally before using it as the input to the B update; the normalized input and the accumulated state B are distinct from raw G. The nucleotide symbol G used in the DNA alphabet is also unrelated. The paper uses the boundary-field label for regions selected by strong local spatial variation. The code's (B) is an accumulated state rather than an instantaneous boundary detector. At the reported horizon it highlights boundary-like bands, while longer audit runs show the thresholded mask spreading across the grid; it should not be read as a permanent stable biological boundary. It is updated through the following thresholded rule:

$$B_{t+1} = B_t + \eta_B S(G - \theta_B)(1 - B_t). \tag{50}$$

Eye-like commitment. The eye-like commitment field is updated only where neural commitment, boundary stabilization, and a positional window are simultaneously active:

$$E_{t+1} = E_t + \eta_E S(N - \theta_{EN}) S(B - \theta_{EB}) A(x, y)(1 - E_t). \tag{51}$$

The thresholding function is

$$S(z) = \frac{1}{1 + e^{-kz}}. \tag{52}$$

This S is a symbolic threshold family in the paper. The execution source implements separate general and eye-specific sigmoid functions with steepness constants 8 and 20, respectively, and clips their exponential arguments; these implementations are not one identical function instance.

Paper form and finite execution. The displayed M and I relations are continuous model forms, while the current code represents them by explicit finite updates with a prescribed source profile. Within each loop, the code first updates M, then updates I using that newly updated M. It next updates and clips N, computes the globally normalized gradient input from the updated I, updates and clips B, and finally updates and clips E using the current N and B, the explicit positional window, and the eye-specific sigmoid. The code applies `DT` to the M and I updates but not to the source-specific N, B, or E accumulation rules. This ordering is an execution property, not a new equation or a simultaneous-update claim.

11.4 Interpretation

The model does not encode a completed neural- or eye-like output array. It does specify localized sources σ_M, diffusion and decay parameters, threshold conditions, response rates, and an explicit positional window for E. Under those supplied constraints, a hierarchical model pattern appears:

1. a smooth morphogen gradient forms in M,
2. an information-density landscape develops in I,
3. a neural-plate-like model region appears through thresholding in N,
4. a boundary-like accumulated pattern is selected from gradient-sensitive input in B,
5. an eye-like localized model pattern appears through secondary thresholding and the positional window in E.

11.5 Key Observation

The emergence proceeds through successive constraints:

$$M \to I \to N \to B \to E. \tag{53}$$

This compact sequence is a staged navigation summary, not the exact dependency graph of the executable update. In the current code, the prescribed source updates M; the newly updated M feeds I; I feeds N and separately supplies the raw gradient that is normalized before updating B; and the updated N and B, together with the positional window, feed E. In particular, N is not a direct input to the B update, so its position before B records staging rather than an N-to-B causal edge. Each stage introduces a higher-order structural condition:

- I: scalar field,
- ∇I: first-order structure,
- thresholding: discretization,
- combined conditions: localization.

11.6 Conceptual Implication

The simulation illustrates that the completed model pattern need not be encoded as an input array. At the reported finite horizon it appears reproducibly within a constrained landscape that includes prescribed source geometry and an explicit positional window; this is not geometry-free emergence, a stationary solution, or an asymptotically stable solution claim.

In this sense, DNA-like parameters act as conditions that shape the geometry of possible states.

11.7 Relation to Bio-IFGT

Within the Bio-IFGT framework:

- I serves as the toy proxy for Bio-IFGT information density,
- $\Phi \sim -I$ defines an effective potential landscape,
- $\mathbf{J} \sim -\nabla\Phi$ induces flows toward stable regions.

These Φ and $\mathbf{J}$ statements are formal or interpretive correspondences; the finite code does not implement a literal one-to-one mapping to independently integrated Φ or $\mathbf{J}$ arrays. None of the toy fields is operationally mapped here to a measured biological observable. The reported finite-horizon structures admit an attractor-like interpretation. The toy run does not establish a formal basin, invariant region, asymptotic convergence, or membership in Ω_{life}.

Each stage in the generation path $M \to I \to N \to B \to E$, read as the staged navigation summary above, can be interpreted as a successive localization of quasi-closure: the morphogen field establishes a global information gradient, which drives local accumulation of I, which in turn crosses thresholds that label persistent sub-regions at the reported horizon. The eye-like commitment E represents a secondary quasi-closure, supported by but distinct from the neural-plate-like region N. This interpretation does not replace the distinct execution dependencies or supply a formal quasi-closure map. The hierarchical structure mirrors the attractor-within-attractor interpretation of differentiation developed in Section 7.7.

11.8 Limitations

This model is intentionally minimal and does not attempt to reproduce biological detail. It provides a finite model-local witness of organization and qualitative structural analogy under constrained dynamics. Longer audit runs continue to evolve, and the limited relaxation and recommitment observed under selected perturbations do not establish general return, biological robustness, or regeneration. No external biological dataset, calibration, or operational model-to-observable map is used here.

12 Simulation Results and Visualization

This section reports a numerical run of the minimal developmental model introduced in the previous section. The aim is to show how structure appears over time in the toy system. The result is not intended as a reproduction of a specific biological process.

12.1 Initial State

At the initial time, the fields are nearly uniform and no organized structure is present. The morphogen-like model field M has only a localized source term. The model constraint-density field I is low and has no meaningful spatial bias apart from a small perturbation. The neural-like commitment field N and the eye-like commitment field E are initially uncommitted model states.

In this state, no completed output array is supplied. The source profile and positional window nevertheless preload coarse spatial information used by the later updates.

12.2 Time Evolution

As the system evolves, structure appears in the following order:

1. the morphogen field diffuses and forms a smooth gradient,
2. the model constraint-density field I accumulates and develops spatial bias,
3. thresholding produces a central neural-plate-like region in N,
4. a globally normalized input derived from $|\nabla I|$ updates the accumulated boundary field B,

5. the overlap of neural commitment, boundary stabilization, and a positional window produces localized eye-like commitment in E.

This list reports the staged presentation of the run. The actual loop is ordered: updated M feeds I, updated I feeds both N and the normalized gradient input to B, and updated N and B combine with the positional window in the E update. It does not contain a direct N-to-B input or a simultaneous-update assumption. The manuscript describes the field process continuously, while the reported run is its source-specific finite-step implementation. No stage contains an encoded final drawing of the resulting structures.

12.3 Final State

At the reported 800-step horizon, the simulation shows a reproducible central neural-plate-like model region and two relatively bright eye-like lobes. The central E values are not zero, and the $E > 0.5$ mask forms one connected component in the audited default condition; this output is therefore not evidence for two independent eye structures. Localization is conditional on the explicit positional window, and disabling the eye pathway suppresses E. This is a finite-horizon description: longer audit runs continue to evolve, and neither stationarity nor asymptotic stability is established.

Thus, the dynamics and supplied spatial constraints jointly shape the model's relative intensity pattern; the result is not geometry-free emergence or a biological eye-development mechanism.

12.4 Figure Descriptions

Each figure corresponds to a stage in the paper's presentation order; that order is a navigation summary rather than a complete execution-dependency graph.

The plotted fields are model states. Printed means, maxima, and fractions above 0.5 are run diagnostics; contours and RGB normalization are visualization transforms. The normalized gradient quantity is different: it is a model intermediate that directly feeds the B update and is not display-only. None of these diagnostics or display transforms is, by itself, an independently measured biological observable.

12.5 Time-Evolution Visualization

An earlier manuscript referenced the following animation path, which is absent from the current distribution:

`simulations/minimal_neural_field/outputs/development_v2_hierarchical.gif`

A historical local animation, `fig8_development_v2_hierarchical.gif`, is retained. Its lineage relative to the currently audited run and tracked PNGs remains unresolved. It is not treated as a verified current supplementary output; no current-run GIF is supplied here.

Independently of that historical animation, the staged model description in the preceding sections is summarized by

$$M \to I \to N \to B \to E. \tag{54}$$

Here the sequence means staged reading order only. The executable dependencies branch from I: one branch updates N, while another forms the normalized gradient input that updates B; E then uses N, B, and the positional window. The displayed arrow sequence does not assert a direct N-to-B causal dependency.

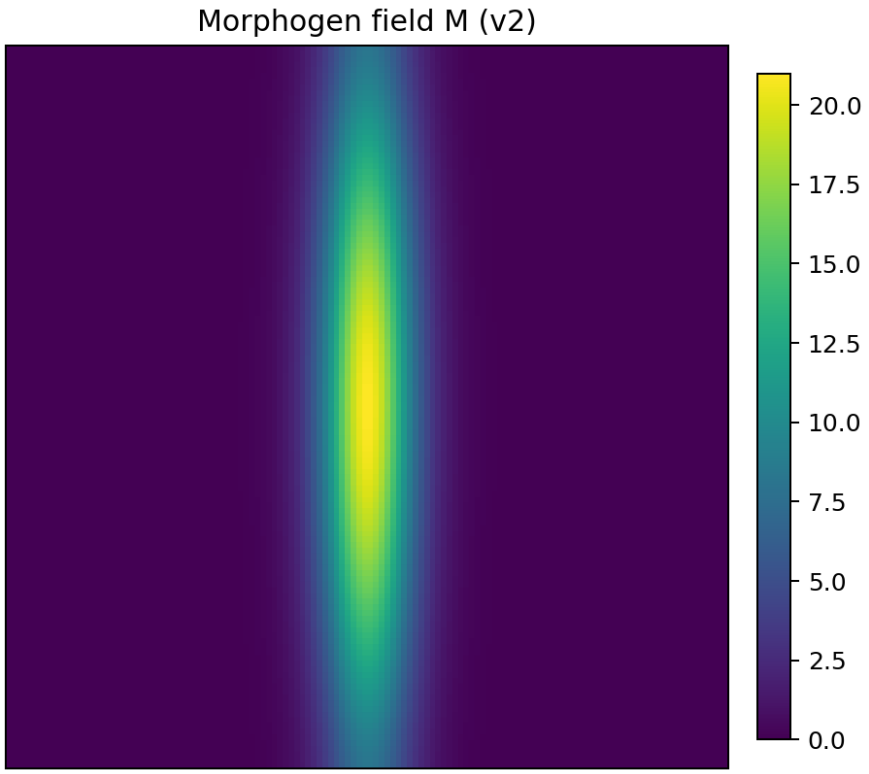


Figure 1: Final morphogen field M. A smooth field forms around the central source.

12.6 Summary of Observations

The simulation illustrates the following points:

- the neural-plate-like model region arises through thresholding of model constraint density,
- the eye-like field localizes through the combination of a gradient-sensitive accumulated state and explicit positional conditions,
- structure appears in stages,
- the completed output array is not specified in advance, although coarse source and window geometry is supplied.

The resulting structures are therefore interpreted as reproducible finite-horizon patterns under constrained dynamics. The run does not establish a stationary solution, invariant basin, or general perturbation return.

12.7 Role of This Section

This result is not a model of a specific biological phenomenon. The role of the simulation is to show finite spatial organization under the stated conditions in a field-based toy system. In this respect, the output is a qualitative structural analogue consistent with the Bio-IFGT interpretation developed in the preceding sections.

It does not demonstrate biological mechanism or empirical correspondence, but only model-local structural possibility. No external biological dataset or operational observable mapping is used.

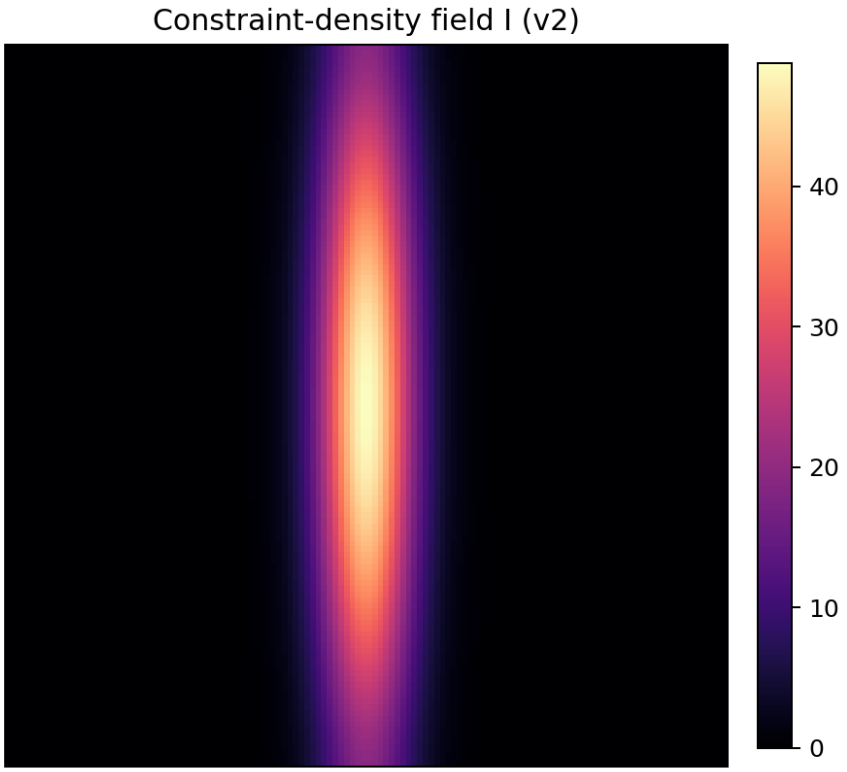


Figure 2: Final model constraint-density field I. The field accumulates in response to M and forms a central peak; no biological measurement map is implied.

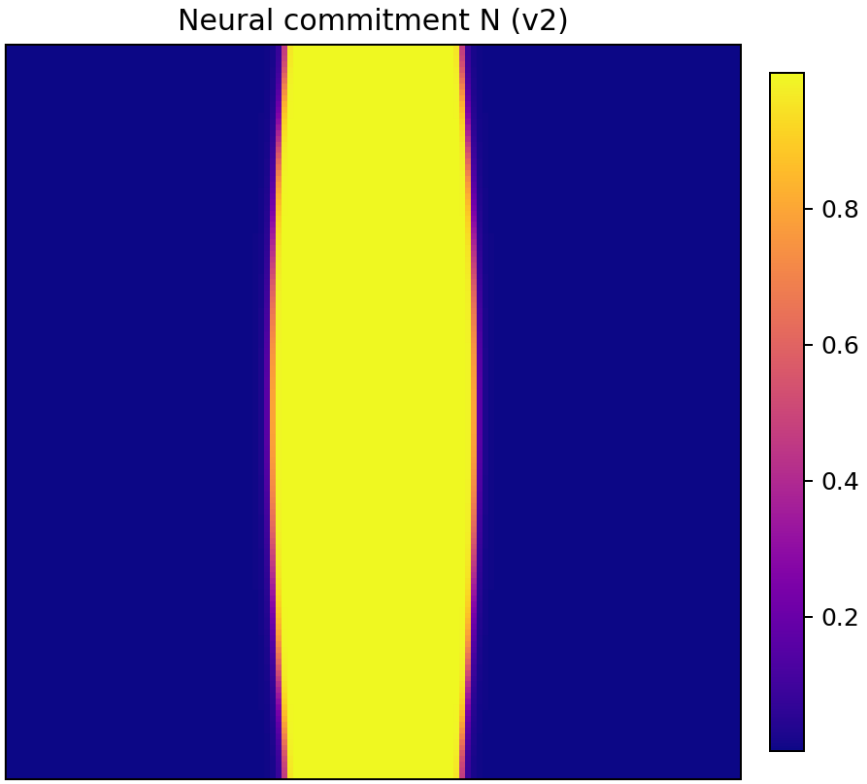


Figure 3: Final neural-like commitment field N. Thresholding of the model constraint density produces a central neural-plate-like model region.

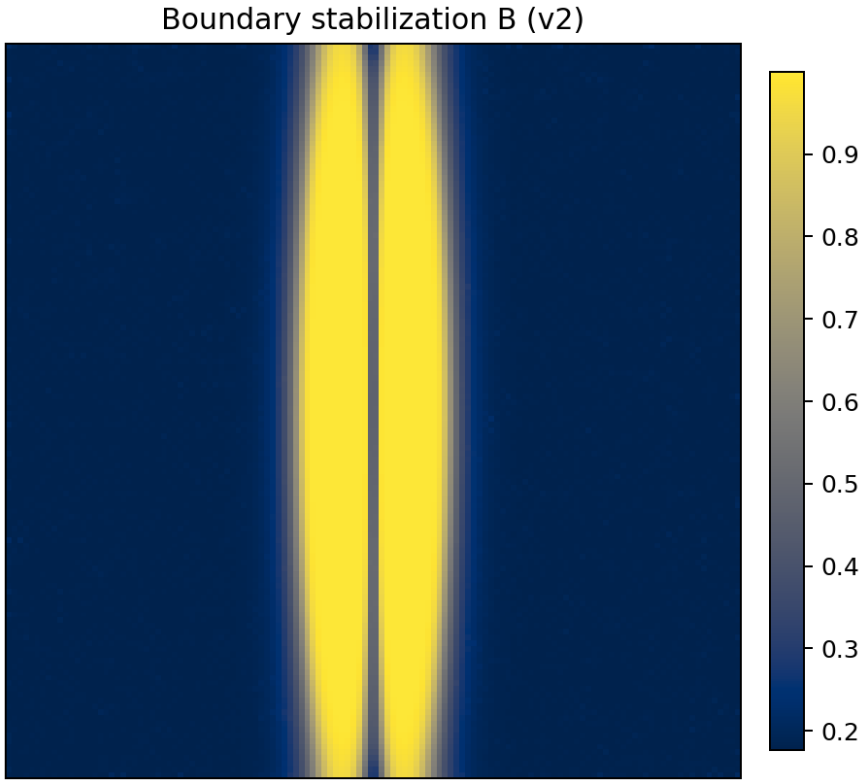


Figure 4: Final accumulated boundary stabilization field B. It is updated from the code's normalized gradient-sensitive input derived from I, highlighting boundary-like regions at the reported horizon; this finite-horizon pattern is not a permanent biological boundary.

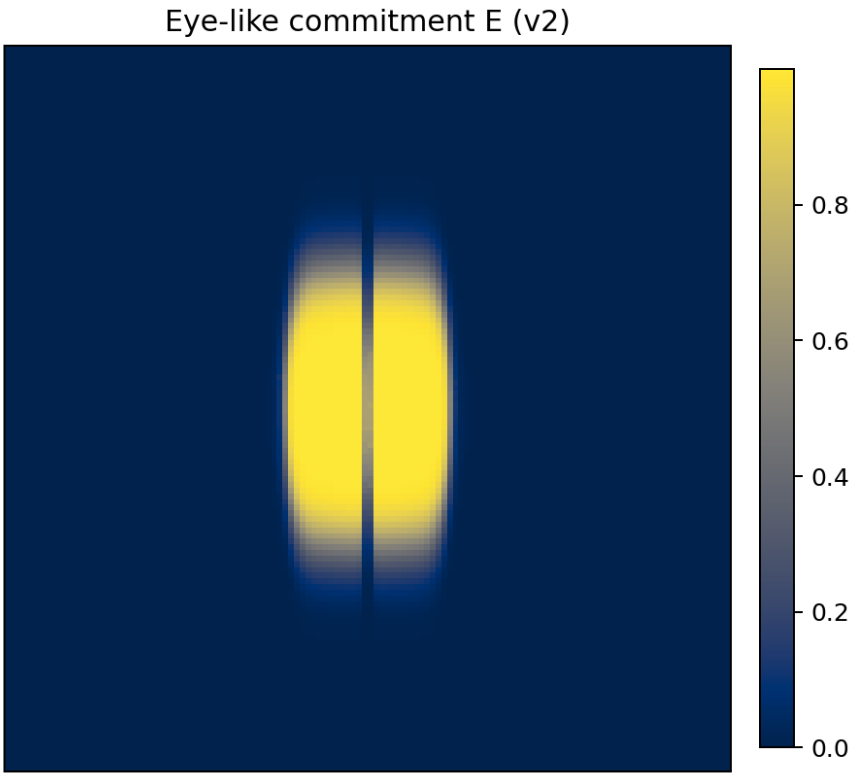


Figure 5: Final eye-like commitment field E. Relative bright regions appear where model commitments overlap within the explicit positional window; this is a qualitative model pattern, not a biological eye identification.

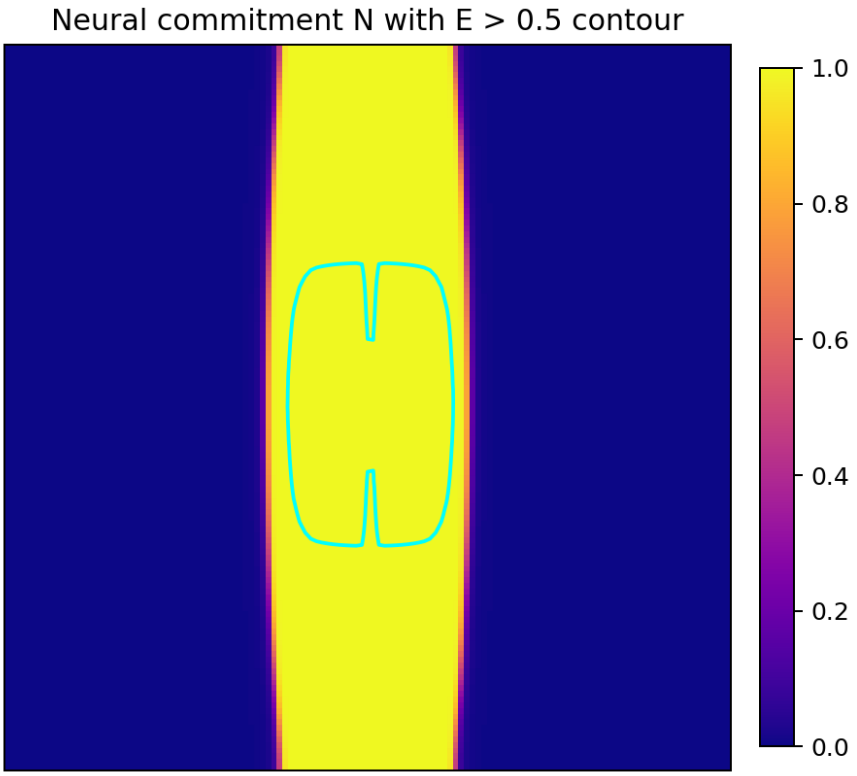


Figure 6: Overlay visualization. The neural commitment field N is shown as the background, with the thresholded eye-like commitment E overlaid as a contour.

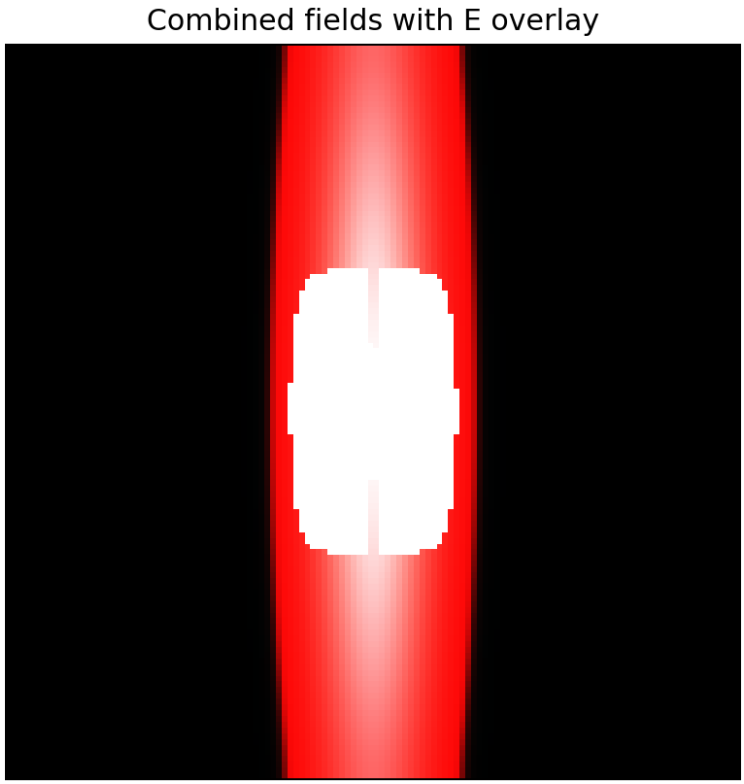


Figure 7: Combined visualization with eye-like commitment. The image summarizes the final relationship among the model fields M, I, N, and E.

13 Discussion

This section summarizes the scope and limits of the framework. The aim is not to add new claims, but to clarify what the framework does and does not address.

13.1 Scope of Description

The paper describes biological systems using source-local information density I, information flow $\mathbf{J}$, and induced potential Φ. This is a minimal common language for treating development, morphogenesis, life maintenance, and the conditions for intelligence; it does not make the quantities operational biological observables or identical carriers across scales.

The framework extracts structure. It does not attempt detailed reproduction of each individual biological phenomenon. Its attractor, maintenance, and robustness language is conditional and interpretive; existence, stability, operational life criteria, and biological regeneration remain open.

13.2 Relation to Existing Models

The framework does not replace existing biological models. Reaction–diffusion systems, gene networks, and neural dynamics remain necessary at their own levels. The present formalism proposes a way to redescribe such models in terms of analogous structural roles: density, flow, induced bias, and stability. Reused vocabulary does not establish identical dynamics or a typed cross-scale map.

It does not require the assumption of a new mechanism, nor does it deny existing theories.

13.3 Non-Closure and Sustained Dynamics

The dynamics considered here do not generally converge to complete equilibrium or fixed closure. Flow continues, and structure is updated. In this sense, biological systems are interpreted as sustained non-closed structures.

This property is not developed as a separate theory in this paper. It appears as a descriptive feature of the dynamics considered.

This interpretation is proposed as structurally analogous to the quasi-closure vocabulary formalized in IFGT: biological systems may be described through regimes where closure is partial and ordinary complete closure is not the target. This is not an identity, derivation, or typed IFGT/Bio translation, and it does not complete any upstream VED/IFGT bridge. The statement does not itself establish a biological maintenance mechanism or the relevant region's existence.

13.4 Status of Time

The paper uses external time t and introduces internal time τ as an interpretive variable. This separates observed time from the progress of internal state update. A precise operational definition of τ is outside the scope of the present paper.

13.5 Limits

Several limits should be stated. The framework does not derive the parameter set θ for a concrete organism. The explicit forms of I, $\mathbf{J}$, and Φ are system-dependent, and no operational biological measurement map is supplied. The paper does not provide numerical predictions or test its toy outputs against an external biological dataset; cited literature provides background rather than a paired model–data comparison.

The framework is therefore a proposed common description, not a detailed or empirically calibrated model of a particular biological system.

13.6 Future Directions

Future work should identify I, $\mathbf{J}$, and Φ in concrete systems, analyze the structure of parameter spaces, and quantify attractor regions. For intelligence, further work must address memory, temporal integration, language, and interaction among individuals.

13.7 Summary

The framework describes biological phenomena in a single formal language without using design or purpose as primitives. It does not address all mechanistic detail or provide numerical prediction. It is best understood as a basis for description and later specialization.

14 Conclusion

This paper described a minimal framework for biological systems using information density I, information flow $\mathbf{J}$, and induced potential Φ. Within this framework, development, life, and intelligence are organized by a proposed attractor-based vocabulary. Concrete convergence, biological life identification, maintenance, stability, regeneration, and second-attractor realization remain separate open questions.

The main point is that biological form can be described without treating design or purpose as primitive explanatory terms. The paper does not aim to reproduce every biological mechanism in detail. It proposes a common descriptive form across biological phenomena; no identity of carriers, dynamics, or scales follows from that reuse.

The framework uses a small set of elements: state, flow, induced bias, and stability. These elements allow development, morphogenesis, life maintenance, probability revision, and the later emergence of intelligence to be placed in one descriptive form.

Several questions remain open. A concrete genotype-to-θ map and parameter provenance must be supplied for particular systems. Operational biological maps, units, and calibration for I, $\mathbf{J}$, and Φ remain unspecified, and the present toy simulation is not compared with an external biological dataset. Typed IFGT/Bio and cross-scale translations, as well as the implementation of intelligence, require additional work.

The result of the paper is therefore not intended as a final theory. It is a form of description. Its usefulness depends on future applications and refinements.